

Sylvie Campagne



European Institute
for Energy Research
by EDF and KIT

Introduction to systematic searching

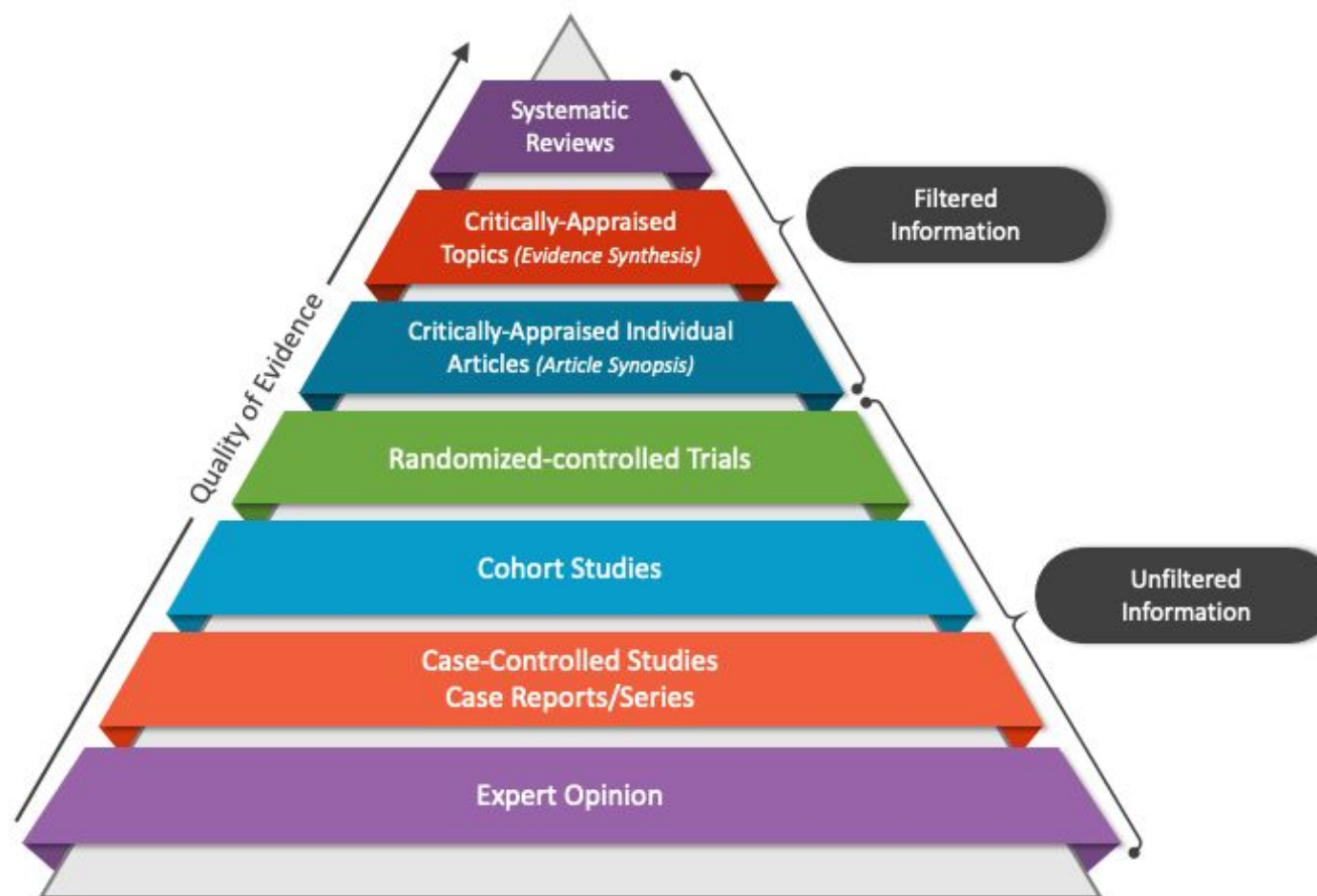
Formation FRB-CESAB 5th to 9th October 2026

Levels of evidence

What is Evidence?

“The available body of facts or information indicating whether a belief or proposition is true or valid.”
Oxford English Dictionary

Generated by scientific studies, which are referred to as “primary” research



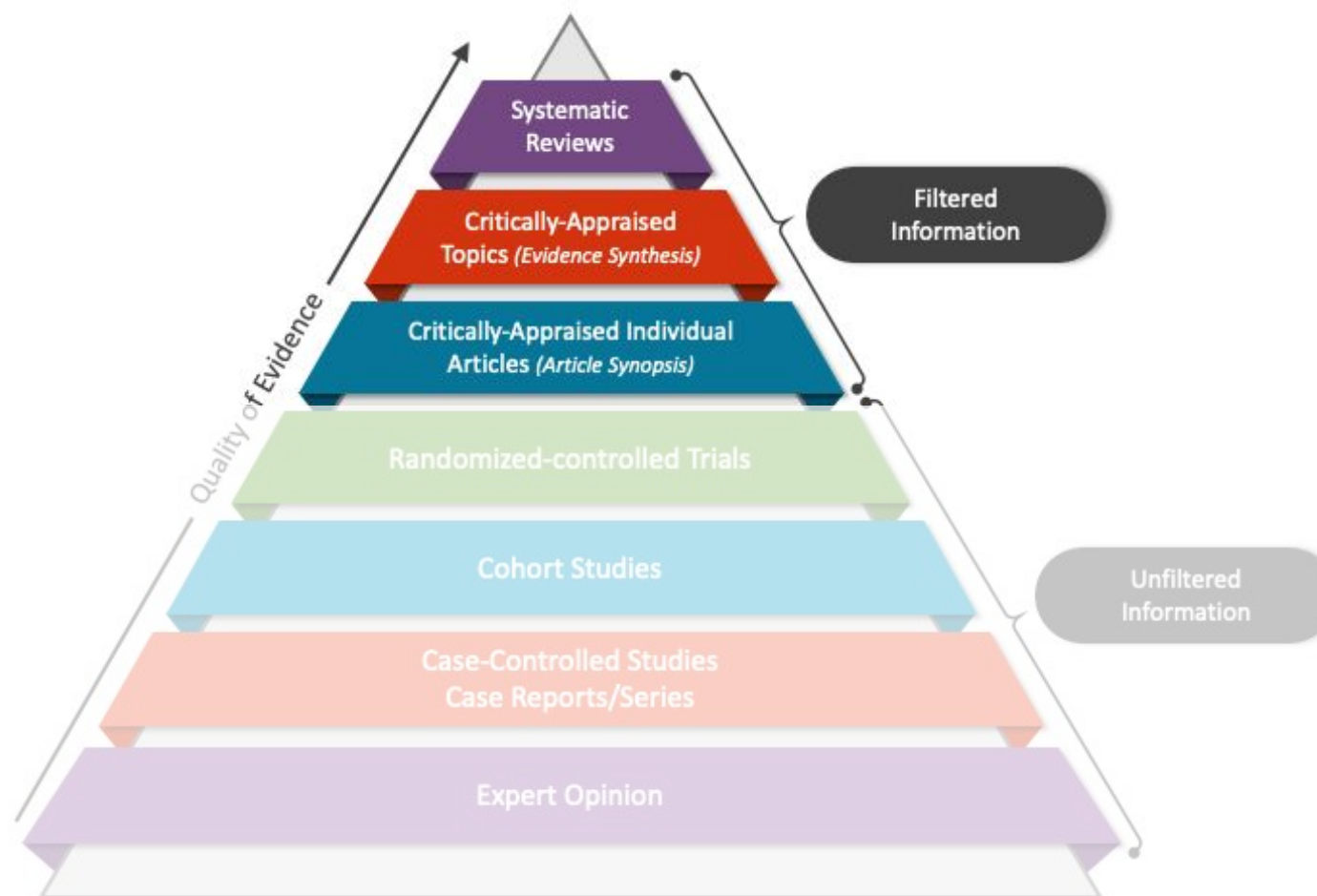
Hierarchy of evidence
(MEDICOWESOME)

Levels of evidence

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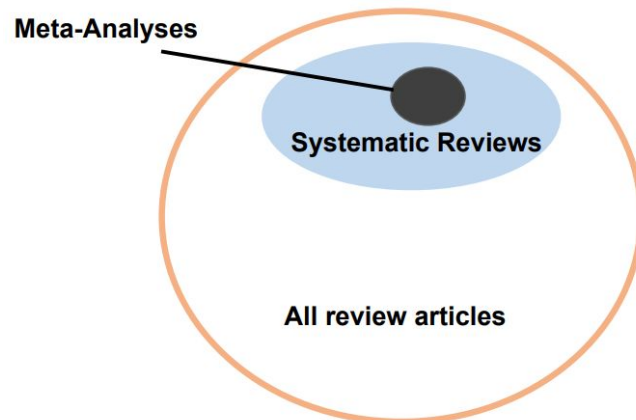


Literature review VS systematic review

Knowledge or evidence synthesis

What is a Systematic Review?

“an evidence synthesis method that aims to answer a specific question as precisely as possible in an unbiased way. The method collates, critically appraises, and synthesizes all available evidence relevant to the question. Reviewers use pre-defined methods to identify risks of bias in the evidence itself, and to minimise bias in the way evidence is identified and selected, and thus provide reliable findings that could inform decision making” (CEE, 2018)



	Systematic Review	Literature Review
Definition	High-level overview of primary research on a focused question that identifies, selects, synthesizes, and appraises all high quality research evidence relevant to that question.	Qualitatively summarizes evidence on a topic using informal or subjective methods to collect and interpret studies.
Goals	Answer a focused clinical question Eliminate bias	Provide summary or overview of topic
Question	Clearly defined and answerable clinical question Recommend using PICO as a guide	Can be a general topic or a specific question
Components	Pre-specified eligibility criteria Systematic search strategy Assessment of the validity of findings Interpretation and presentation of results Reference list	Introduction Methods Discussion Conclusion Reference list
Number of Authors	Three or more	One or more
Timeline	Months to years Average eighteen months	Weeks to months
Requirements	Thorough knowledge of topic Perform searches of all relevant databases Statistical analysis resources (for meta-analysis)	Understanding of topic Perform searches of one or more databases
Value	Connects practicing clinicians to high quality evidence Supports evidence-based practice	Provides summary of literature on a topic

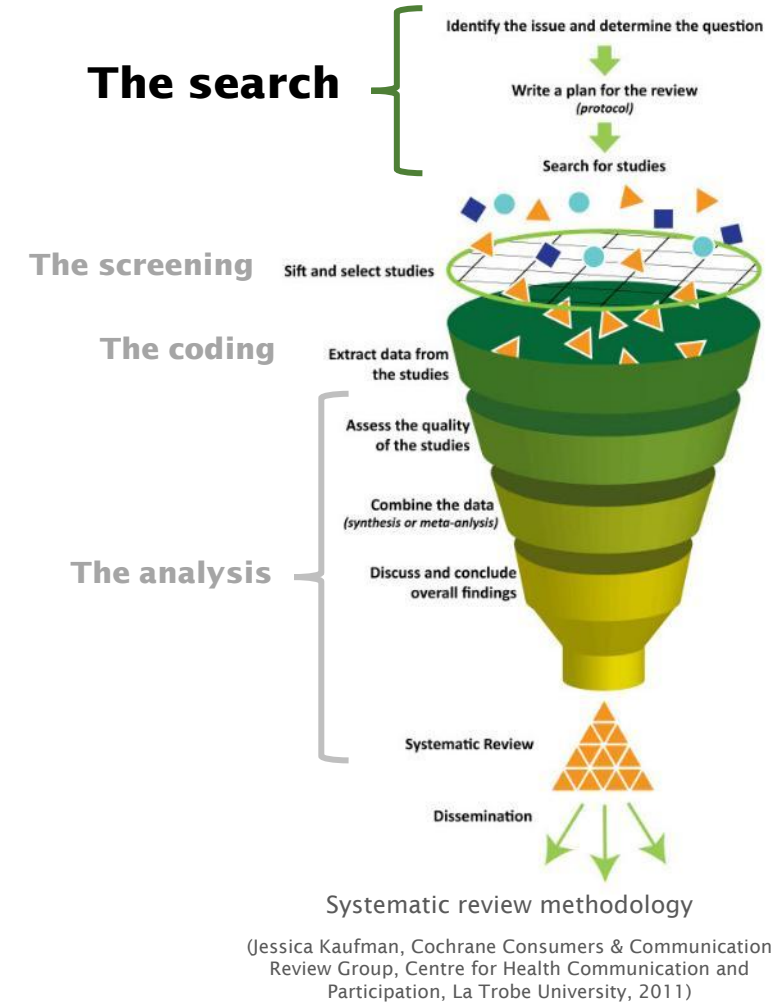
Systematic review and literature review

(Kysh, Lynn (2013): Difference between a systematic review and a literature review. figshare. Poster.<https://doi.org/10.6084/m9.figshare.766364.v1>)

The search phase

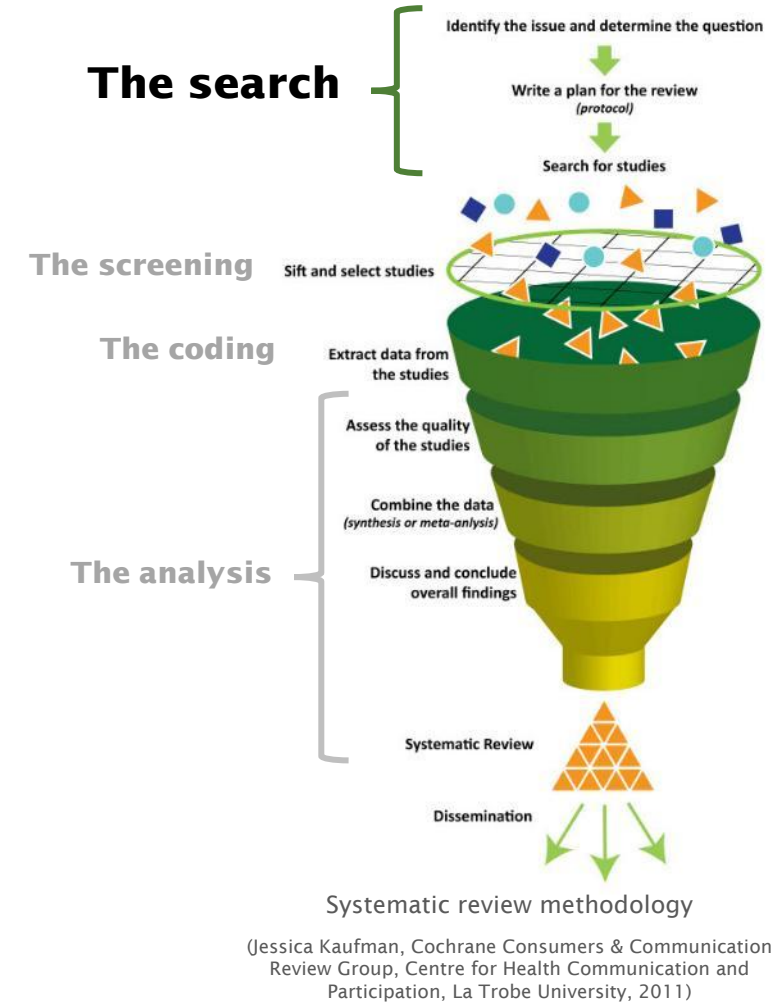
The search phase

Search the references related to our subject or question



The search phase : 4 steps

- Define the question
- Define the search terms
- Define the search string
- Choose the search platform



The search phase : 4 steps

- Define the question →
- Define the search terms →
- Define the search string
- Choose the search platform

What are the ecosystem services provided by wetlands ecosystems?

The keywords chosen are: service + ecosystem + environmental + wetland + peatland + marsh etc.

The search

Identify the issue and determine the question
↓
Write a plan for the review (protocol)
↓
Search for studies

The screening

Sift and select studies

The coding

Extract data from the studies

Assess the quality of the studies

Combine the data (synthesis or meta-analysis)

The analysis

Discuss and conclude overall findings

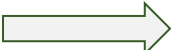
Systematic Review

Dissemination

Systematic review methodology

(Jessica Kaufman, Cochrane Consumers & Communication Review Group, Centre for Health Communication and Participation, La Trobe University, 2011)

The search phase : 4 steps

- Define the question  *What are the ecosystem services provided by wetlands ecosystems?*
- Define the search terms  *The keywords chosen are: service + ecosystem + environmental + wetland + peatland + marsh etc.*
- Define the search string  *TS = ((ecosystem* OR environment*) AND service*)
AND TI = ("wetland" AND "peatland" AND "marsh" ...).*
- Choose the search platform  *Web of Science (WoS) Core Collection
Google Scholar*

The search



The screening

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Systematic Review

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	SCOPUS	Web of Science
Search operators (AND, OR, *, \$)	https://blog.scopus.com/posts/boolean-searches-in-scopus-understanding-operator-precedence-best-practices	https://webofscience.help.clarivate.com/en-us/Content/search-operators.html
Where do we search?	ALL AUTHOR-NAME TITLE-ABS-KEY	TS=Topic TI=Title AB=Abstract

The search phase : many steps

Search the references related to our subject or question

1. PLANNING THE SEARCH

- 1.1 Establishing a test-list
- 1.2 Identifying search terms
- 1.3 Identifying relevant sources of articles
- 1.4 Choosing bibliographic management software
- 1.5 Addressing the need for grey literature
- 1.6 Deciding when to stop
- 1.7 Submitting the search strategy in the protocol for peer-review

2. CONDUCTING THE SEARCH

- 2.1 Prioritizing bibliographic sources
- 2.2 Building the search string
- 2.3 Assessing retrieval performance
- 2.4 Refining the results
- 2.5 Searching for grey literature
- 2.6 Additional approaches

3. MANAGING REFERENCES AND REPORTING

- 3.1 Keeping track of the search and recording results
- 3.2 Writing the search report

4. UPDATING AND AMENDING A SEARCH

A guide to the planning, conduct, management and reporting of the searching phase of systematic reviews and systematic maps

Livoreil, B., Glanville, J., Haddaway, N.R. et al. (2017). Systematic searching for environmental evidence using multiple tools and sources. *Environ Evid* 6, 23 <https://doi.org/10.1186/s13750-017-0099-6>

The search

Identify the issue and determine the question
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Write a plan for the review (protocol)
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Combine the data (synthesis or meta-analysis)

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Systematic Review

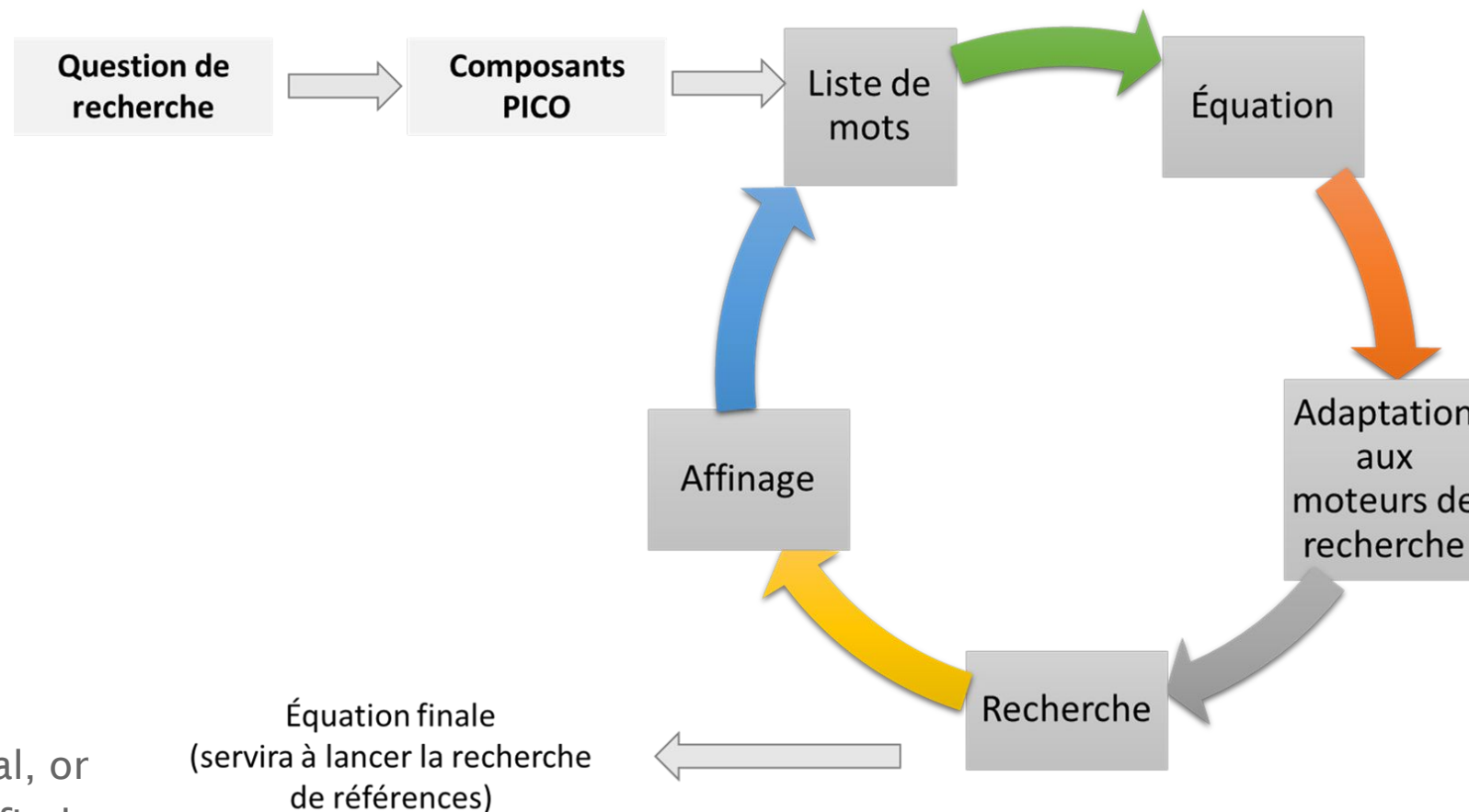
Dissemination

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The search strategy

- Define the question
- Define the search terms
- Define the search string



search terms encompasses individual, or compound words used in a search to find relevant articles

search string is a combination of search terms combined using Boolean operators

The search string

PICO/PECO elements (Richardson et al. 1995)

Population : effect *on what?*

Intervention / Exposure : effect *of what ?*

Comparator : *compared to what? to what reference?*

Outcome : effect *measured by what?*

(**Context** : *what type of study?*)

Question element	Definition
Population (of subjects)	Unit of study (e.g. ecosystem, species) that should be defined in terms of the statistical populations of subject(s) to which the intervention will be applied.
Intervention/exposure	Proposed management regime, policy, action or the environmental variable to which the subject populations are exposed.
Comparator	Either a control with no intervention/exposure or an alternative intervention or a counterfactual scenario.
Outcome	All relevant outcomes from the proposed intervention or environmental exposure that can be reliably measured

The search string

1

Define the PICO based on the research question

PICO/PECO elements (Richardson et al. 1995)

My PICO

Population : effect *on what?*

Intervention / Exposure : effect *of what ?*

Comparator : *compared to what? to what reference?*

Outcome : effect *measured by what?*

(**Context** : *what type of study?*)



Example with the Agri-TE project:

What is the effect of agricultural practices on biodiversity at the global level?

The search string

1

Define the PICO based on the research question

PICO/PECO elements (Richardson et al. 1995)

Population : effect *on what?*

Intervention / Exposure : effect *of what ?*

Comparator : *compared to what? to what reference?*

Outcome : effect *measured by what?*

(**Context** : *what type of study?*)



My PICO

Any unplanned/uncultivated taxon

Any agricultural practice

Agricultural witness or natural environment of ref.

Effect-size representing a biodiv metric.

Meta-analyses only

Example with the Agri-TE project:

What is the effect of agricultural practices on biodiversity at the global level?

The search string

2

Establish the list of words that will be used to construct the search equation

My search terms

biodiversity, soil fauna, birds, butterflies

tillage, fertilization, pesticides

croplands, forest

species richness, biomass, Shannon's entertainment

meta-analyses



My PICO

Any unplanned/uncultivated taxon

Any agricultural practice

Agricultural witness or natural environment of ref.

Effect-size representing a biodiv metric.

Meta-analyses only

Example with the Agri-TE project:

What is the effect of agricultural practices on biodiversity at the global level?

The search string

3

Building the search string by adapting to search engines (eg: WoS)

My search terms

biodiversity, soil fauna, birds, butterflies

tillage, fertilization, pesticides

croplands, forest

species richness, biomass, Shannon's diversity

meta-analyses



My search string

TS= ((biodiversity OR soil fauna OR birds OR butterflies)

AND (tillage OR fertilizers **OR** pesticides)

AND (croplands OR forest)

AND (species richness OR biomass OR Shannon's diversity)

AND (meta-analyses))

Example with the Agri-TE project:

What is the effect of agricultural practices on biodiversity at the global level?

The search string

My search string

TS= ((biodiversity OR soil fauna OR birds OR butterflies)

AND (tillage OR fertilizers **OR** pesticides)

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AND (species richness OR biomass OR Shannon's diversity)

AND (meta-analyses))

Topic (title, abstract,
keywords, keywords plus)

Logical and Boolean
Operators

Both depend on
the bibliographic
sources

Example with the Agri-TE project:

What is the effect of agricultural practices on biodiversity at the global level?

The search string

My search string

TS= ((biodiversity OR soil fauna OR birds OR butterflies)
AND (tillage OR fertilizers **OR** pesticides)
AND (croplands OR forest)
AND (species richness OR biomass OR Shannon's diversity)
AND (meta-analyses))

Exact Expression
"soil fauna"

Truncations
pesticide*, pesticide\$

Exclusion
NOT (medical science OR
economics)

Thematic
soil fauna OR (earthworms
OR spiders OR collembola OR
springtails)

Beware of database variations in the search equation!!!

- Some use a different language for searching
- For example, \$ instead of *.
- Additional options (inside or nearby)
- Help files are useful!
- Check the options
- Seek specialist help if necessary
- SAVE EVERYTHING

The search string

4

Test the search string

My search string

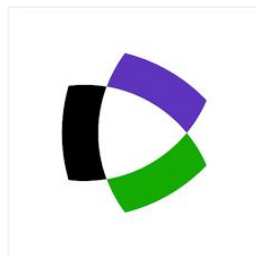
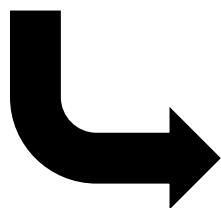
TS= ((biodiversity OR soil fauna OR birds OR butterflies)

AND (tillage OR fertilizers **OR** pesticides)

AND (croplands OR forest)

AND (species richness OR biomass OR Shannon's diversity)

AND (meta-analyses))



200 results is not enough!
20,000 results is too much!
Refinement needed...

The search string

My search string

TS= ((biodiversity OR soil fauna OR birds OR butterflies)
AND (tillage OR fertilizers **OR** pesticides)
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AND (meta-analyses))

5

Refine the search string

Iterative process that can (must?) be long

Ex: Foo et al. (2021)

Initial search string	1	TS = ((("terminal investment" OR "reproductive effort" OR "fecundity compensation") AND ("immune challeng*" OR "immunochalleng*" OR "infect*")) NOT (load OR human OR people))
159 results		
Add inclusion terms	2	TS = ((("terminal investment" OR "reproductive effort" OR "fecundity compensation" OR "reproductive compensation" OR "fitness") AND ("immune challeng*" OR "immunochalleng*" OR "infect*" OR lipopolysaccharide OR lps OR phytohemagglutinin OR pha OR "sheep red blood cells" OR srbc OR implant OR vaccin*)) NOT (load OR human OR people))
4,360 results		
Edit inclusion term	3	TS = ((("terminal investment" OR "reproductive effort" OR "fecundity compensation" OR "reproductive compensation" OR "reproductive fitness") AND ("immune challeng*" OR "immunochalleng*" OR "infect*" OR lipopolysaccharide OR lps OR phytohemagglutinin OR pha OR "sheep red blood cells" OR srbc OR implant OR vaccin*)) NOT (load OR human OR people))
493 results		
Add inclusion terms	4	TS = ((("terminal investment" OR "reproductive effort" OR "fecundity compensation" OR "reproductive compensation" OR "reproductive fitness" OR "reproductive investment" OR "Life History Trade-Off*" OR "life history") AND ("immune challeng*" OR "immunochalleng*" OR "infect*" OR lipopolysaccharide OR lps OR phytohemagglutinin OR pha OR "sheep red blood cells" OR srbc OR implant OR vaccin*)) NOT (load OR human OR people))
2,489 results		
Change inclusion term	5	TS = ((("terminal investment" OR "reproductive effort" OR "fecundity compensation" OR "reproductive compensation" OR "reproductive fitness" OR "reproductive investment" OR "Life History Trade-Off*" OR "life history" OR "trade off") AND ("immune challeng*" OR "immunochalleng*" OR "infect*" OR lipopolysaccharide OR lps OR phytohemagglutinin OR pha OR "sheep red blood cells" OR srbc OR implant OR vaccin*)) NOT (load OR human OR people))
1,819 results		
Delete inclusion term	6	TS = ((("terminal investment" OR "reproductive effort" OR "fecundity compensation" OR "reproductive compensation" OR "reproductive fitness" OR "reproductive investment" OR "reproductive success" OR "Life History Trade-Off*" OR "trade off") AND ("immune challeng*" OR "immunochalleng*" OR "infect*" OR lipopolysaccharide OR lps OR phytohemagglutinin OR pha OR "sheep red blood cells" OR srbc OR implant OR vaccin*)) NOT (load OR human OR people))
1,155 results		
Add inclusion term	7	TS = ((("terminal investment" OR "reproductive effort" OR "fecundity compensation" OR "reproductive compensation" OR "reproductive fitness" OR "reproductive investment" OR "reproductive success" OR "Life History Trade-Off*" OR "Phenotypic Plasticity") AND ("immune challeng*" OR "immunochalleng*" OR "infect*" OR lipopolysaccharide OR lps OR phytohemagglutinin OR pha OR "sheep red blood cells" OR srbc OR implant OR vaccin*)) NOT (load OR human OR people))
1,429 results		
Add exclusion terms	8	TS = ((("terminal investment" OR "reproductive effort" OR "fecundity compensation" OR "reproductive compensation" OR "reproductive fitness" OR "reproductive investment" OR "reproductive success" OR "Life History Trade-Off*" OR "Phenotypic Plasticity") AND ("immune challeng*" OR "immunochalleng*" OR "infect*" OR lipopolysaccharide OR lps OR phytohemagglutinin OR pha OR "sheep red blood cells" OR srbc OR implant OR vaccin*)) NOT (load OR human OR people OR men OR women OR infant* OR rat OR rats OR mouse OR mice OR pig* OR pork OR beef OR cattle OR sheep OR lamb* OR chicken* OR calf* OR horse*))
1,141 results		
Pilot 100 papers to check hit rate, 6% hit rate. Continue refining.		
Final search string		TS = ((("terminal investment" OR "reproductive effort" OR "fecundity compensation" OR "reproductive compensation" OR "reproductive fitness" OR "reproductive investment" OR "reproductive success" OR "Life History Trade-Off*" OR "Phenotypic Plasticity" OR "pre-copulatory NEAR/5 trait*" OR "sexual NEAR/5 weapon*" OR "sexual NEAR/5 ornament*" OR "post-copulatory NEAR/5 trait*" OR "ejaculate quality" OR "sperm quality" OR "mating effort" OR "parental care") AND ("immune challeng*" OR "immunochalleng*" OR "infect*" OR lipopolysaccharide OR lps OR phytohemagglutinin OR pha OR "sheep red blood cells" OR srbc OR implant OR vaccin* OR nylon OR sephadex)) NOT (load OR human OR people OR men OR women OR infant* OR rat OR rats OR mouse OR mice OR pig* OR pork OR beef OR cattle OR sheep OR lamb* OR chicken* OR calf* OR horse* OR infective))
1,567 results (~10% hit rate)		

The search string

My search string

TS= ((biodiversity OR soil fauna OR birds OR butterflies)
AND (tillage OR fertilizers **OR** pesticides)
AND (croplands OR forest)
AND (species richness OR biomass OR Shannon's diversity)
AND (meta-analyses))

5

Refine the search string

Iterative process that can (must?) be long

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Final
search
string

Final search string	1,567 results (~10% hit rate)
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The test list

Test-list : studies that you wish to include in your systematic review and which you know meet the inclusion criteria.

- Discuss the list (involve partners/co-authors/colleagues) to construct it and then consolidate it
- Extract metadata
- Order of magnitude, ca. 30 items

Interest : verify the capacity of a research equation to capture studies corresponding to the aim of our systematic review.

- Calculate the miss rate = the % of items belonging to the test list not captured by the equation

It must be minimized, ie the equation must approach 100% of the captured test-list... Refinement possible.

The test list

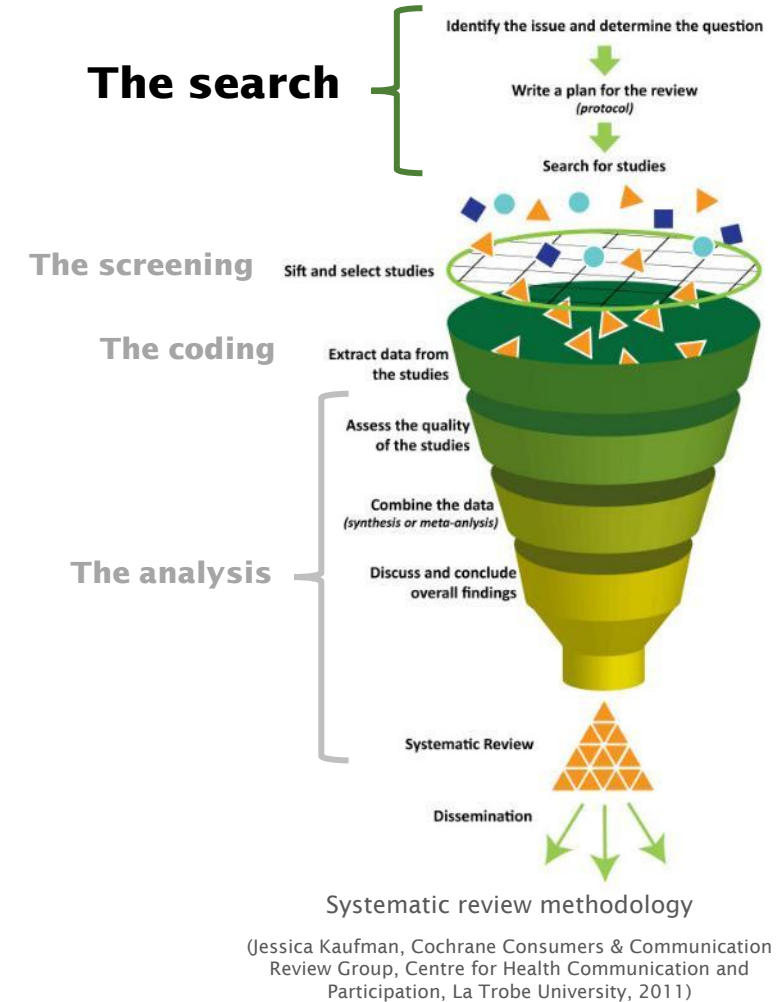
Example of test list

Campagne et al. (2023)

DOI	References		Retrieved by WOS	Retrieved by Scopus	Retrieved by google scholar
10.3389/fev	Belgrano et al.	Mapping and evaluating marine p	1	oui	
10.3389/fma	Cavanagh et	Future risk for Southern Ocean E	2	oui	
10.3354/mep	Cheung, W.V	Application of macroecological th	3	oui	
10.1093/ices	Cheung, W.V	Integrating ecophysiology and pl	4	non	oui
10.1016/j.glo	Cinner et al.	Vulnerability of coastal communi	5	oui	
10.1016/j.eco	Cook et al. (2	Towards marine ecosystem base	6	oui	
10.5670/oce	Cooley et al.	Ocean acidification's potential to	7	oui	
10.1088/174	Cooley, S.R.	Anticipating ocean acidification's	8	oui	
10.1111/gcb.	Fernandes, J.	Modelling the effects of climate c	9	oui	
10.1007/978	Marcos et al.	Reviewing the ecosystem service	10	oui	
10.1002/Ino.	Orcutt et al.	Impacts of deep-sea mining on m	11	oui	
10.1890/070	Palumbi, S.R.	Managing for ocean biodiversity t	12	oui	
10.1007/s11	Roessig et al.	Effects of global climate change c	13	non	non
10.1016/j.jnc	Roncin, N., A	Uses of ecosystem services provi	14	oui	
10.1126/scie	Worm B., E.B	Impacts of biodiversity loss on oc	15	oui	
10.1016/j.o	Kermagoret,	How does eutrophication impact	16	oui	
10.17159/saj	Arabi, S., Na	Impacts of marine plastic on ecos	17	oui	
10.2307/234	Depellegrin,	Integrating ecosystem service val	18	oui	
10.1016/j.eco	Broszeit, S., E	What can indicators of good envi	19	oui	
10.1371/jour	Pendleton, L	Estimating global "Blue Carbon" e	20	oui	
10.1042/ETL	Hall-Spencer	Ocean acidification impacts on co	21	oui	
10.1016/j.ma	Potts, T., Bur	Do marine protected areas delive	22	oui	
10.1016/j.jer	Lemasson, A.	Linking the biological impacts of	23	oui	
10.3389/fma	Pouso, S., Bo	An Interdisciplinary Approach for	24	oui	
10.1016/j.o	Song, J., Zhar	Changes in ecosystem services va	25	oui	
10.1016/j.en	Yim, J., Kwor	Analysis of forty years long chang	26	oui	
http://www.	Hicks, C.C., M	Trade-offs in values assigned to e	27	oui	
10.1016/B97	Leenhardt, P	The Role of Marine Protected Are	28	NOT in WoS	non
10.1007/s10	Selim, S.A., B	Direct and indirect effects of clim	29	NOT in WoS	oui
10.3391/ai.2	Katsanevakis	Impacts of invasive alien species	30	NOT in WoS	oui
			25 out of 30	28 out of 30	29 out of 30
			83,3%	93,3%	96,7%
			Only in WOS		
			25 out of 27		
			92,6%		

The search phase

- Choose the search platform



The bibliographic sources

Bibliographic sources capture any source of references, including electronic bibliographic databases, those sources which would not be classified as databases (e.g. the Internet via search engines), hand searched journals, and personal contacts.

- Bibliographic
 - eg WoS, Scopus, OpenAlex, Pubmed
- Web search tools
 - eg Google, Google Scholar
- Grey literature sources
 - Organizational websites
 - Thesis repositories

Scopus®

PubMed

OpenAlex

WEB OF SCIENCE®



EThOS
ELECTRONIC THESES ONLINE SERVICE
Opening access to UK theses

The bibliographic sources

Bibliographic databases

- Web of Science
- Scopus
- Agricola
- AGRIS (FAO)
- Academic Search Premier
- Biological Abstracts
- CAB Abstracts
- etc.

Example with web of science

WEB OF SCIENCE™



THOMSON REUTERS™

Search

Web of Science™ Core Collection

My Tools

Search History

Marked List

Welcome to the new Web of Science! View a brief tutorial.

Basic Search

Example: oil spill* mediterranean

Topic

▼

Search

Click here for tips to improve your search.

+ Add Another Field

Reset Form

TIMESPAN

All years

From 1945 to 2017

▼ MORE SETTINGS

Web of Science Core Collection: Citation Indexes

☒ Science Citation Index Expanded (SCI-EXPANDED) --1945-present

☒ Social Sciences Citation Index (SSCI) --1956-present

☒ Arts & Humanities Citation Index (A&HCI) --1975-present

☒ Conference Proceedings Citation Index- Science (CPCI-S) --1990-present

☒ Conference Proceedings Citation Index- Social Science & Humanities (CPCI-SSH) --1990-present

☒ Book Citation Index-- Science (BKCI-S) --2005-present

☒ Book Citation Index-- Social Sciences & Humanities (BKCI-SSH) --2005-present

☒ Emerging Sources Citation Index (ESCI) --2015-present

Data last updated: 2017-02-07

Auto-suggest publication names

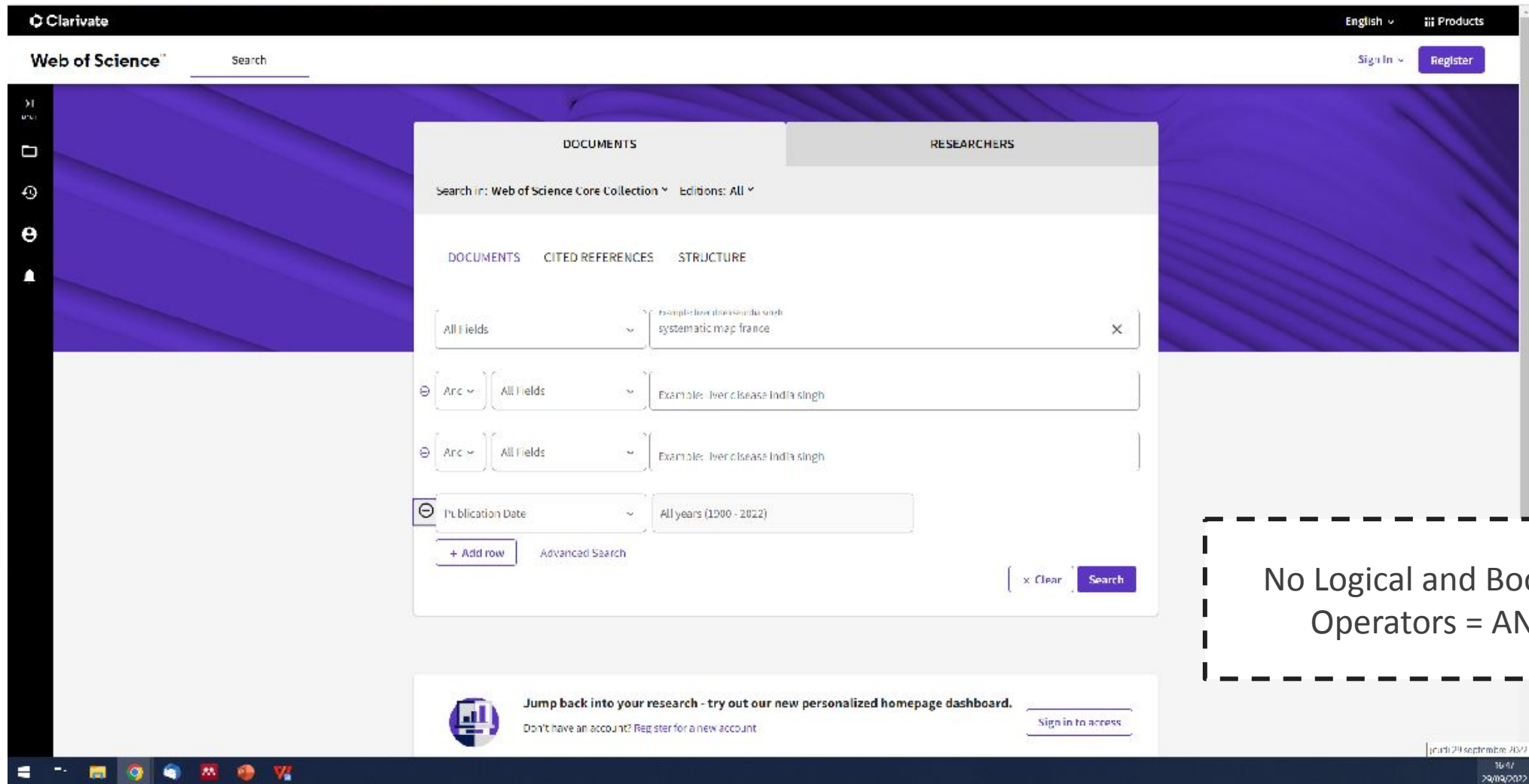
(The Autosuggest service is not available.)

Default Number of Search Fields to Display

1 field (Topic)

Example with web of science

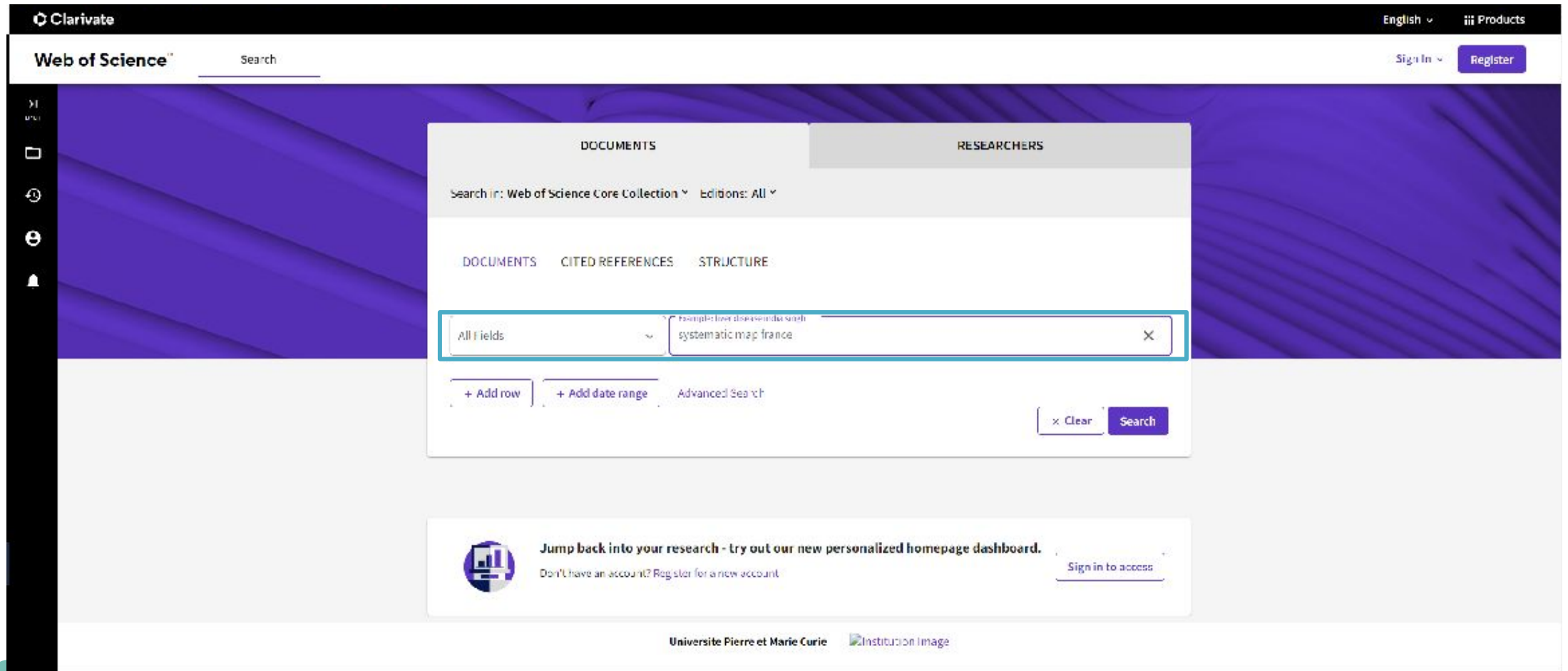
<https://www.webofscience.com/wos/woscc/basic-search>



The screenshot shows the Web of Science search interface. The top navigation bar includes the Clarivate logo, language selection (English), and product links. The main header features the 'Web of Science' logo and a search bar. The search form is divided into two tabs: 'DOCUMENTS' and 'RESEARCHERS'. The 'DOCUMENTS' tab is active, showing a search scope of 'Web of Science Core Collection' and 'Editions: All'. Below this, there are three search criteria rows. The first row has 'All Fields' selected and the text 'Example: liver disease india singh systematic map france'. The second and third rows have 'All Fields' selected and the text 'Example: liver disease india singh'. The fourth row has 'Publication Date' selected and the range 'All years (1900 - 2022)'. There are buttons for '+ Add row', 'Advanced Search', 'x Clear', and 'Search'. At the bottom, there is a promotional banner for a personalized homepage dashboard and a 'Sign in to access' link.

No Logical and Boolean Operators = AND

Example with web of science



The screenshot displays the Clarivate Web of Science search interface. At the top, the Clarivate logo is on the left, and 'English' and 'Products' are on the right. Below this, the 'Web of Science' header includes a search bar and 'Sign In' and 'Register' buttons. A left sidebar contains navigation icons. The main content area features two tabs: 'DOCUMENTS' (selected) and 'RESEARCHERS'. Under the 'DOCUMENTS' tab, there are filters for 'Search in: Web of Science Core Collection' and 'Editions: All'. Below these are three sub-tabs: 'DOCUMENTS', 'CITED REFERENCES', and 'STRUCTURE'. A search input field is highlighted with a red border, containing the text 'systematic map france' and a dropdown menu set to 'All fields'. Below the search field are buttons for '+ Add row', '+ Add date range', and 'Advanced Search'. At the bottom right of the search area are 'Clear' and 'Search' buttons. A promotional banner at the bottom encourages users to 'Jump back into your research - try out our new personalized homepage dashboard' and provides a 'Sign in to access' link. The footer includes the text 'Universite Pierre et Marie Curie' and an 'Institution Image' placeholder.

Clarivate

English Products

Web of Science Search

Sign In Register

DOCUMENTS RESEARCHERS

Search in: Web of Science Core Collection Editions: All

DOCUMENTS CITED REFERENCES STRUCTURE

All fields systematic map france

+ Add row + Add date range Advanced Search

Clear Search

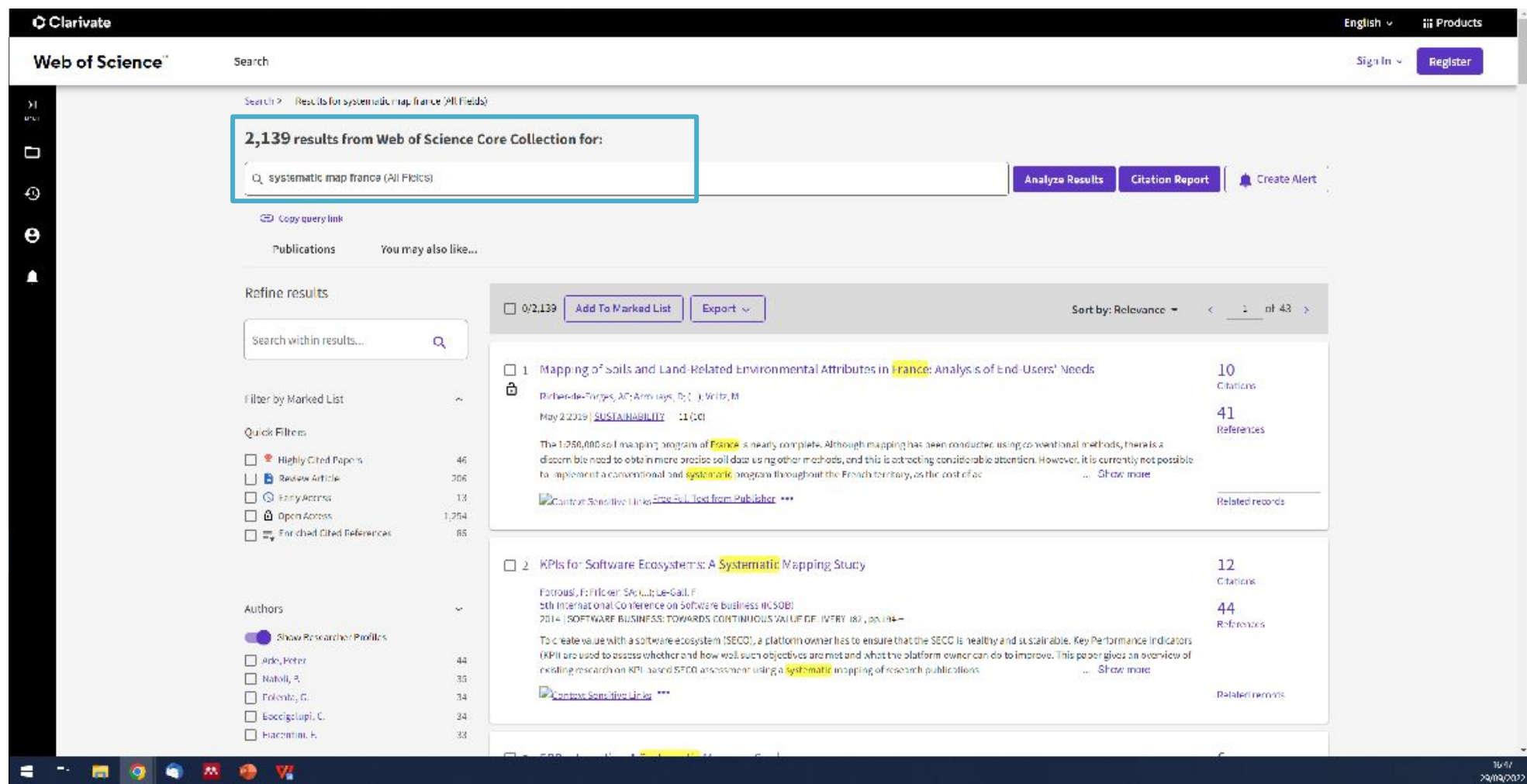
Jump back into your research - try out our new personalized homepage dashboard.

Don't have an account? Register for a new account

Sign in to access

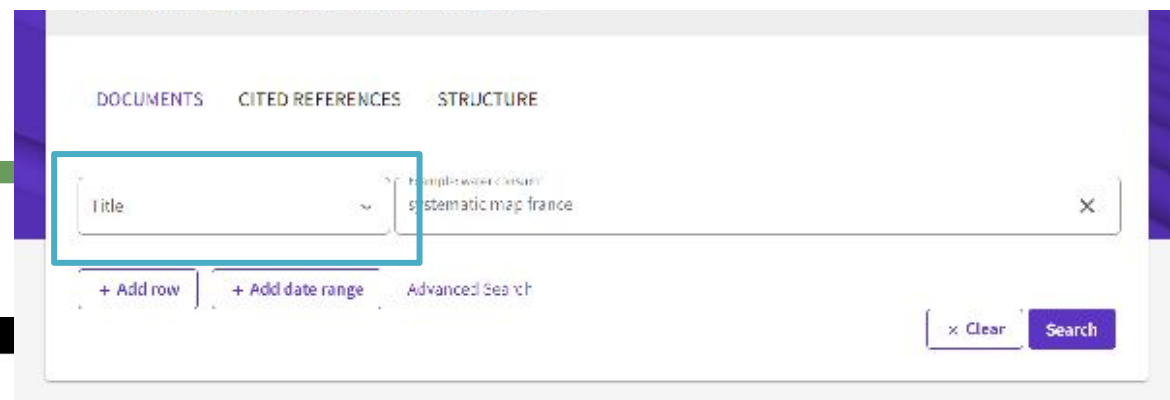
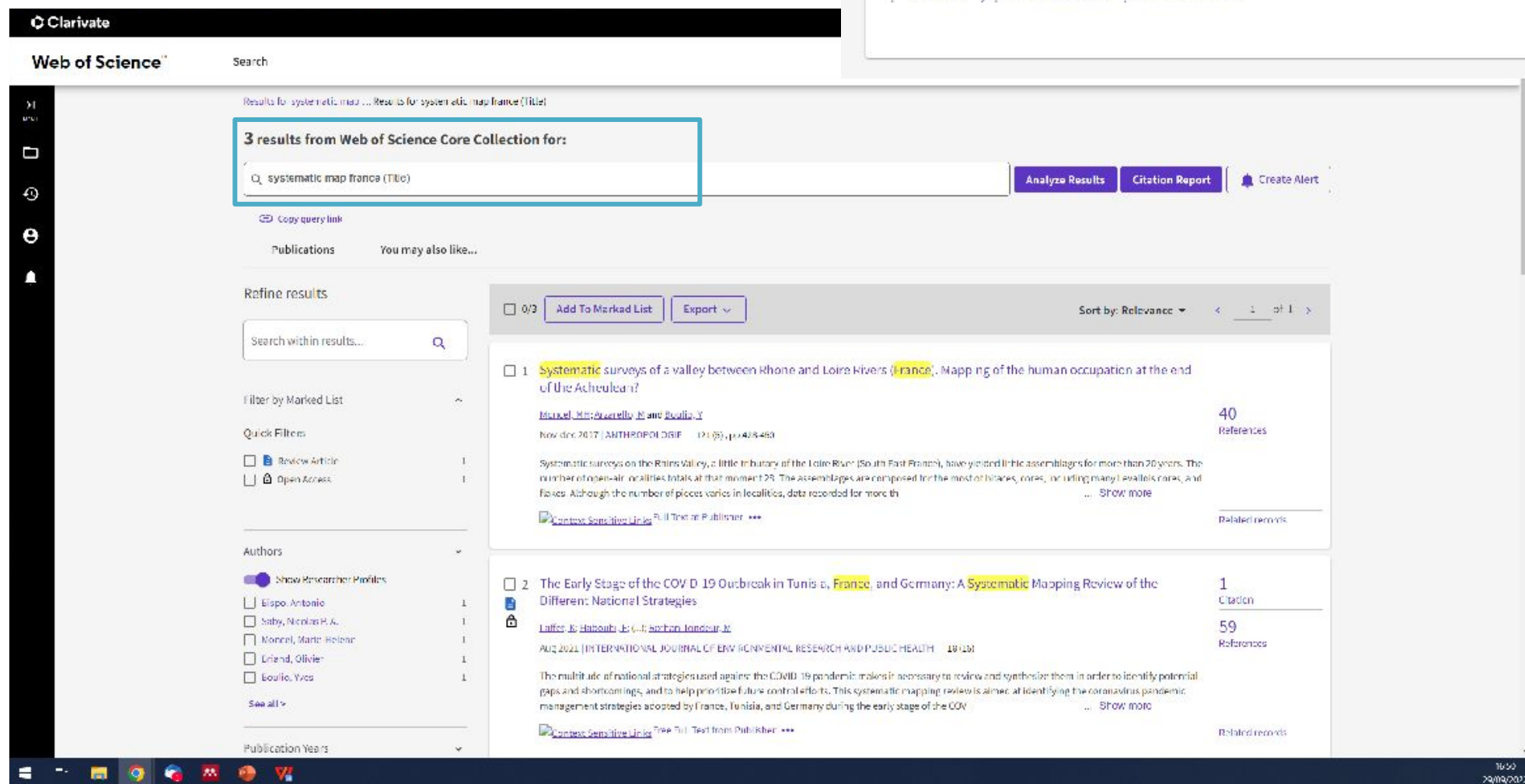
Universite Pierre et Marie Curie Institution Image

Example with web of science



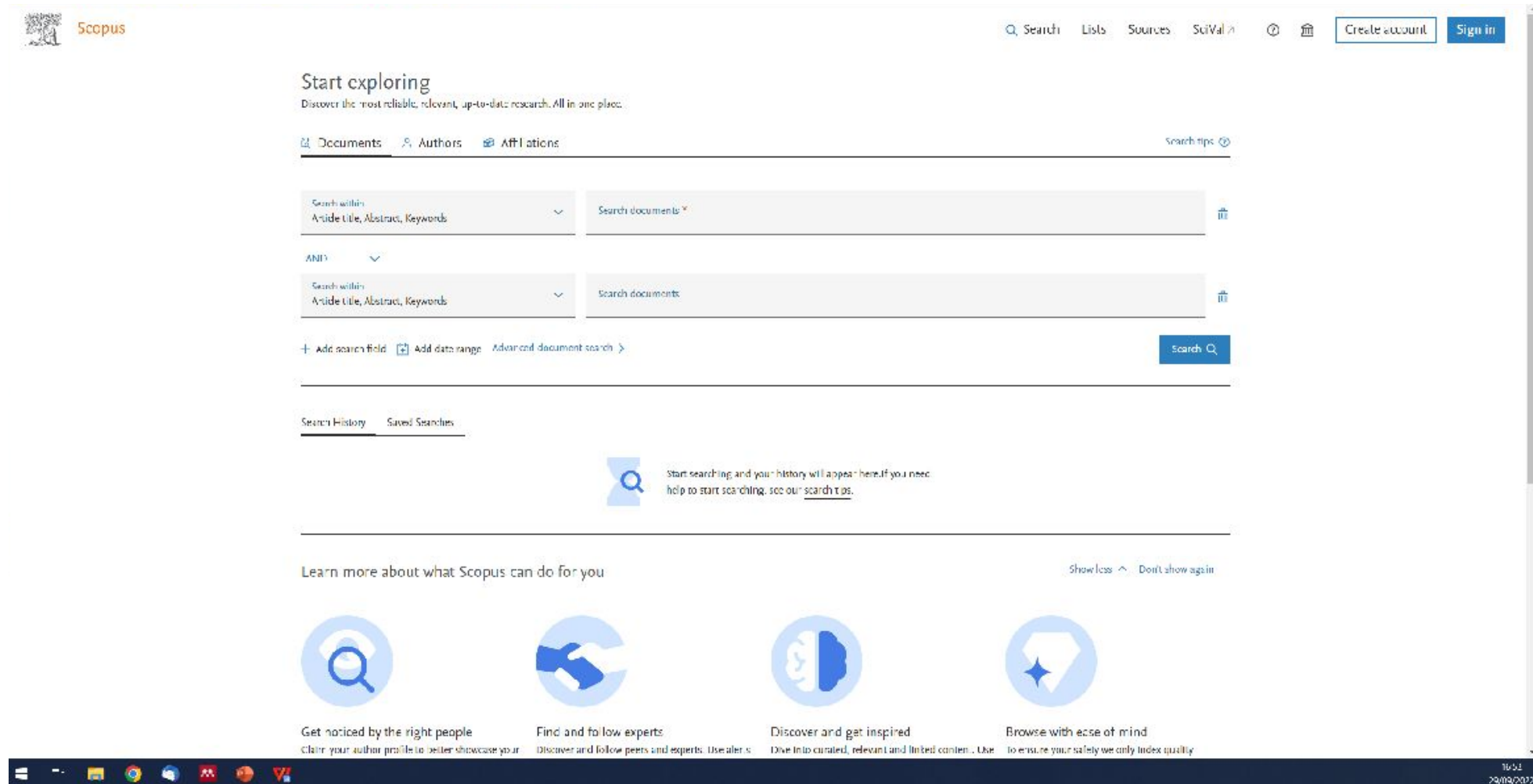
The screenshot displays the Clarivate Web of Science interface. At the top, the navigation bar includes the Clarivate logo, language settings (English), and product links. The main header shows the 'Web of Science' logo and a search bar. Below the search bar, a box highlights the search results: '2,139 results from Web of Science Core Collection for: systematic map france (All Fields)'. To the right of the search bar are buttons for 'Analyze Results', 'Citation Report', and 'Create Alert'. Below the search bar, there are links for 'Copy query link', 'Publications', and 'You may also like...'. The left sidebar contains a 'Refine results' section with a search bar and a 'Filter by Marked List' section. The main content area shows a list of results, with the first two results highlighted. The first result is 'Mapping of soils and Land-Related Environmental Attributes in France: Analysis of End-Users' Needs' by Rothermel-Turner, A.C.; Arriaga, D.; J. Witz, M., published in SUSTAINABILITY in May 2019. The second result is 'KPIs for Software Ecosystems: A Systematic Mapping Study' by Potroulakis, P.; Fritzen, S.; Lutz, U.; Gail, F., published in the 5th International Conference on Software Business (ICSOB) in 2014. The interface also includes a 'Sort by: Relevance' dropdown and a '1 of 43' pagination indicator.

Example with web of science

This screenshot shows the search results page on the Web of Science platform. The header includes the 'Clarivate' logo and the 'Web of Science' title. The search results are for the query 'systematic map france (Title)'. The results are displayed in a list format, with the first two results highlighted. The first result is 'Systematic surveys of a valley between Rhone and Loire Rivers (France). Mapping of the human occupation at the end of the Acheulean?' by Michel M. G. G. and Boule, Y., published in ANTHROPOLOGIE in 2017. The second result is 'The Early Stage of the COVID-19 Outbreak in Tunisia, France, and Germany: A Systematic Mapping Review of the Different National Strategies' by Laif, S., Hamoudi, J., and Amour, J., published in the INTERNATIONAL JOURNAL OF ENVIRONMENTAL RESEARCH AND PUBLIC HEALTH in 2021. The left sidebar contains filters for 'Refine results', 'Filter by Marked List', 'Quick Filters', 'Authors', and 'Publication Years'. The right sidebar shows 'Add To Marked List', 'Export', 'Sort by: Relevance', and 'Page 1 of 1'.

Example with Scopus



The screenshot shows the Scopus homepage. At the top, there's a navigation bar with links for Search, Lists, Sources, SciVal, and user options like Create account and Sign in. The main section is titled "Start exploring" with the tagline "Discover the most reliable, relevant, up-to-date research. All in one place." Below this, there are tabs for Documents, Authors, and Affiliations. A search bar is prominently displayed with a dropdown menu for "Search within" (Article title, Abstract, Keywords) and a "Search documents" button. There are also links for "Add search field", "Add date range", and "Advanced document search". Below the search bar, there's a section for "Search History" and "Saved Searches". At the bottom, there's a "Learn more about what Scopus can do for you" section with four icons and descriptions: "Get noticed by the right people", "Find and follow experts", "Discover and get inspired", and "Browse with ease of mind".

Scopus

Search Lists Sources SciVal Create account Sign in

Start exploring
Discover the most reliable, relevant, up-to-date research. All in one place.

Documents Authors Affiliations Search tips

Search within
Article title, Abstract, Keywords

Search documents

AND

Search within
Article title, Abstract, Keywords

Search documents

+ Add search field + Add date range Advanced document search Search

Search History Saved Searches

Start searching, and your history will appear here. If you need help to start searching, see our [search tips](#).

Learn more about what Scopus can do for you Show less Don't show again

Get noticed by the right people
Claim your author profile to better showcase your work.

Find and follow experts
Discover and follow peers and experts. Use alerts to stay up-to-date.

Discover and get inspired
Dive into curated, relevant and linked content. Use Scopus Insights to explore new research.

Browse with ease of mind
To ensure your safety we only index quality content.

16/01 20/03/2022

The bibliographic sources

Web search tools

- Google
- Ecosia
- Bing
- DuckDuck Go

....

!!! Depend of connection
parameters!!!!

The bibliographic sources

Grey literature sources

- 'File drawer' research / unpublished research results
 - Unfinished/published/accepted articles
 - The theses
 - The “uninteresting” results
 - Non-academic studies
 - Technical reports
 - Government documents
 - Internal reports
- all results not intended for academic publication

The bibliographic sources

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- Non-academic studies

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 - Government documents
 - Internal reports
- all results not intended for academic publication

How to find them?

- *Calls for evidence (social media, networks)*
- *Thesis databases (eg eThOS)*
 - *Google Scholar, Google*
- *Pre-print servers (eg ArchivX)*
 - *Organizational websites*

Example with Publish or Perish



Harzing's Publish or Perish (Windows GUI Edition) 8.2.3883.8074

File Edit Search View Help

My searches

Trash

Search terms	Source	Papers	Cites	Cites/year	h	g	hI, norm	hI, annual	hA	acc 10
(marine OR coastal OR ocean) A...	Google Scho...	500	132703	5529.29	155	361	92	3.83	55	278

No search selected

Select an existing search to inspect or modify it, or click one of these buttons to create a new search.

Crossref*

OpenAlex*

Google Scholar*

Google Scholar Profile*

PubMed*

Scopus**

Semantic Scholar***

Web of Science***

Import External Data...

About importing external data

* Free data source

** Free registration required

*** External subscription

Citation metrics

Help

Publication years:
Citation years:
Papers:
Citations:
Cites/year:
Cites/paper:
Authors/paper:
h-index:
g-index:
hI, norm:
hI, annual:
hA-index:
Papers with ACC >= 1,2,5,10,20:

Copy Results

Save Results

Paper details

Help

Select a paper in the results list
(to the left of this pane) to see
its details here.

Copy Paper Details

Tools

Preferences...

Online User's Manual

Frequently Asked Questions

Training Resources

YouTube Channel

Become a PoP Supporter

Cites	Per year	Rank	Authors	Title	Year	Publication	Publisher

Example with Publish or Perish



Harzing's Publish or Perish (Windows GUI Edition) 8.2.3883.8074

File Edit Search View Help

My searches

Trash

Search terms	Source	Papers	Cites	Cites/year	h	g	hI, norm	hI, annual	hA	acc10
(marine OR coastal OR ocean) A...	Google Scho...	500	132703	5529.29	155	361	92	3.83	55	278

Google Scholar search

Authors: Years:

Publication name: ISSN:

Title words:

Keywords: (marine OR coastal OR ocean) AND (species OR biodiversity OR ecosystem) AND "ecosystem services" AND char

Maximum number of results: Include: ☒ CITATION records ☒ Patents

Citation metrics [Help](#)

Publication years: 1997-2021
Citation years: 24 (1997-2021)
Papers: 500
Citations: 132703
Cites/year: 5529.29
Cites/paper: 265.41
Authors/paper: 3.65
h-index: 155
g-index: 361
hI, norm: 92
hI, annual: 3.83
hA-index: 55
Papers with ACC >= 1,2,5,10,20:
493,476,396,278,168

Copy Results

Save Results

Tools

Preferences...

Online User's Manual

Frequently Asked Questions

Training Resources

YouTube Channel

Become a PoP Supporter

Cites	Per year	Rank	Authors	Title	Year	Publication	Publis
✓ h 184	15.33	1	SR Cooley, HL Kite...	Ocean acidification's potential to ...	2009	Oceanography	STOP
✓ h 417	34.75	2	H Mooney, A Lang...	Biodiversity, climate change, and e...	2009	Current opinion in ...	Elsev
✓ h 413	19.67	3	CM Duarte	Marine biodiversity and ecosystem...	2000	Journal of experimental m...	Elsev
✓ 51	4.64	4	KMA Chan, M Ruck...	Characterizing changes in marine ...	2010	F1000 biology reports	ncbi
✓ h 200	11.33	5	C Hattam, JP Atkin...	Marine ecosystem services: linking...	2015	Ecological ...	Elsev
✓ 56	9.33	6	RK Turner, M Scha...	Coastal zones ecosystem services	2015	Valuation of Ecosystem Ser...	Spring
✓ h 272	22.67	7	SR Palumbi, PA San...	Managing for ocean biodiversity b...	2009	Frontiers in Ecology ...	Wiley
✓ h 319	25.00	8	EF Grarek, S Polask...	Ecosystem services as a common L...	2010	Conservation ...	Wiley

Paper details [Help](#)

Select a paper in the results list (to the left of this pane) to see its details here.

Copy Paper Details

Example of search strings depending on the bibliographic sources

	Name	Search field	Search string	Search hits	Date of search (DD/MM/YYYY)
LITERATURE DATABASES	Web of science	TS	((marine OR coast* OR ocean OR sea OR littoral OR maritime) AND (species OR biodiversity OR ecosystem OR ecological) AND ("ecosystem service\$" OR "contribution to people" OR "ecosystem function\$" OR "ecosystem process" OR "landscape service\$" OR "disservice\$" OR "provisioning service\$" OR ((provision OR production OR exploitation) AND (food OR fisher* OR macroalgae\$ OR molecules)) OR "biomass for nutrition" OR "biomass for materials" OR "genetic materials" OR "raw materials" OR "maintain* food webs" OR "life cycle maintenance and habitat protection" OR "habitat provision" OR "nursery function" OR "regulation service\$" OR "climate regulation" OR "carbon sequestration" OR "weather regulation" OR "atmospheric composition and conditions" OR "air quality regulation" OR "coastal protection" OR "water retention" OR "nutrient regulation" OR "nutrient cycling" OR "pathogen regulation" OR "pest and disease control" OR "mediation of waste" OR "mediation of mass" OR "cultural service\$" OR "intellectual interaction" OR "physical interaction" OR "experiential interaction\$" OR tourism OR recreation OR amenity OR aesthetic OR heritage OR symbolic OR "cognitive effect\$" OR "knowledge production" OR education) AND (dynamic\$ OR impact\$ OR effect\$ OR variation\$ OR interaction\$ OR evolution OR change\$)).	17329	20/07/2021
	Scopus	TITLE-ABS-KEY	((marine OR coast* OR ocean OR sea OR littoral OR maritime) AND (species OR biodiversity OR ecosystem OR ecological) AND ("ecosystem service\$" OR "contribution to people" OR "ecosystem function\$" OR "ecosystem process" OR "landscape service\$" OR "disservice\$" OR "provisioning service\$" OR ((provision OR production OR exploitation) AND (food OR fisher* OR macroalgae\$ OR molecules)) OR "biomass for nutrition" OR "biomass for materials" OR "genetic materials" OR "raw materials" OR "maintain* food webs" OR "life cycle maintenance and habitat protection" OR "habitat provision" OR "nursery function" OR "regulation service\$" OR "climate regulation" OR "carbon sequestration" OR "weather regulation" OR "atmospheric composition and conditions" OR "air quality regulation" OR "coastal protection" OR "water retention" OR "nutrient regulation" OR "nutrient cycling" OR "pathogen regulation" OR "pest and disease control" OR "mediation of waste" OR "mediation of mass" OR "cultural service\$" OR "intellectual interaction" OR "physical interaction" OR "experiential interaction\$" OR tourism OR recreation OR amenity OR aesthetic OR heritage OR symbolic OR "cognitive effect\$" OR "knowledge production" OR education) AND (dynamic\$ OR impact\$ OR effect\$ OR variation\$ OR interaction\$ OR evolution OR change\$)).	24051	20/07/2021
ONLINE SEARCH ENGINE	Google Scholar	keywords	(marine OR coastal OR ocean) AND (species OR biodiversity OR ecosystem) AND "ecosystem services" AND change	300	22/07/2021
ORGANIZATIONAL WEBSITES	FAO	Language: "English"	fishery	50	27/08/2021
	UNESCO	Filter: language: "English" - source: "UNESCO" - AuthoCorporate-en-s: "Intergovernmental Oceanographic Commission" - nature of content: "guide" AND "manuals and handbooks"	marine ecosystem service	50	19/08/2021
	UNEP	Filters: "Reports and publications" AND "Publication" AND "Report", "Ecosystems and biodiversity" AND "oceans and seas"	marine ecosystem service	50	19/08/2021
	US NOAA		ecosystem service	15	19/08/2021
	EEA		marine ecosystem service	7	19/08/2021
	IUCN		ecosystem service	32	27/08/2021

The search strategy

Complementary measures of the efficiency of the equation

- **Miss-rate** : thanks to the test-list, must be minimized
- **Hit-rate** : Percentage of relevant articles, calculated on a sample (for example, on 100 randomly selected results)
→ aim **for at least 10%**
- **Number of results** : depending on the specifications of the search questions and the tool (IA ?) used during the screening stage

Adaptation depends on the search engine used and/or the strategy employed.



European Institute
for Energy Research
by EDF and KIT

Thank You



+49 (0) 721 6105 1330



sylvie.campagne@eifer.org



www.eifer.org